https://docs.espressif.com/projects/esp-idf/en/v5.1.5/esp32s3/api-reference/peripherals/spi_master.html

GPIO Matrix and IO_MUX

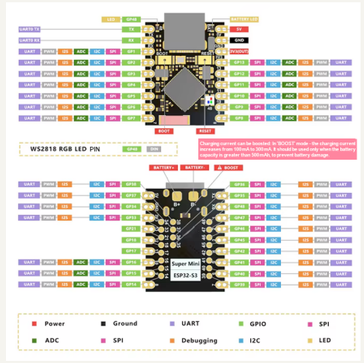
Most of chip's peripheral signals have direct connection to their dedicated IO_MUX pins. However, the signals can also be routed to any other available pins using the less direct GPIO matrix. If at least one signal is routed through the GPIO matrix, then all signals will be routed through it.

When an SPI Host is set to 80MHz or lower frequencies, routing SPI pins via GPIO matrix will behave the same comparing to routing them via IOMUX.

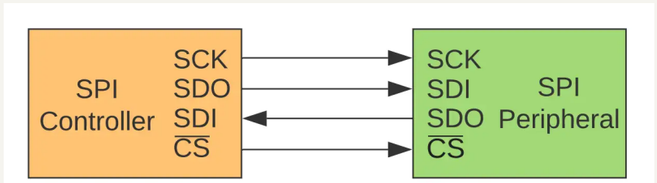
The IO_MUX pins for SPI buses are given below.

Pin Name	GPIO Number (SPI2)
CS0 [1]	10
SCLK	12
MISO	13
MOSI	11
QUADWP	14
QUADHD	9

[1] : Only the first Device attached to the bus can use the CS0 pin.



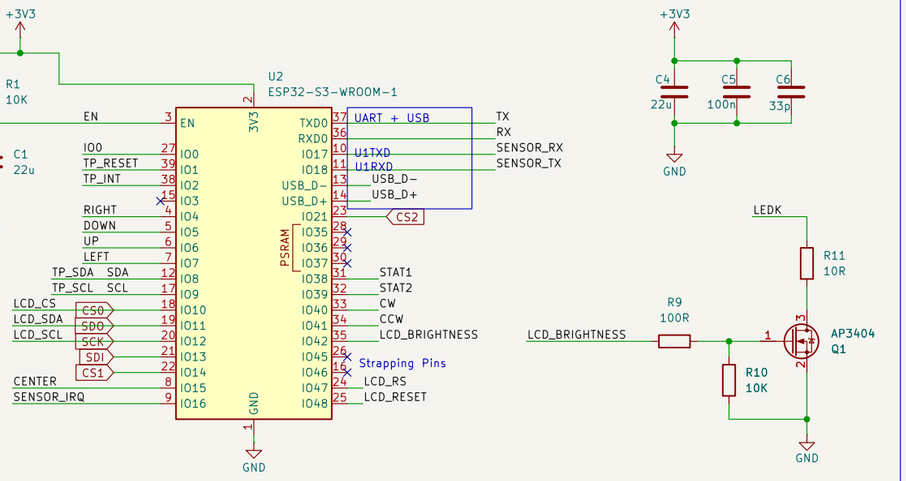
https://oshwa.org/resources/a-resolution-to-redefine-spi-signal-names/



Mirror Vertically (Y)
As the Pinout is Flipped in the LCD Connector.

https://yarsa.tech/files/DT80160S096NA-2310BS001.pdf

ESP32-S3

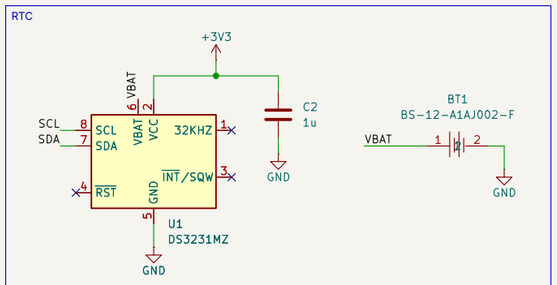


8.2.2.3 Recommended Passive Components

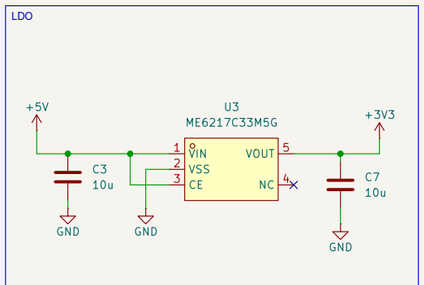
Low ESR ceramic capacitors, such as X7R or X5R, are preferred for input decoupling capacitors and should be placed as close as possible to the supply and ground pins of the IC. Due to voltage derating of the capacitors, it is recommended that 25V rated capacitors are used for the IN and SYS pins, which normally operate at 5V. After derating, the minimum capacitance must be greater than 1μF.

		MIN	NOM	MAX	UNIT
C_{SYS}	Capacitance on SYS pin	1	10	100	μF
C_{BAT}	Capacitance on BAT pin	1	1	-	μF
C_{IN}	Capacitance on IN pin ($R_{ILIM_PRESENT}$ > 25ms)	1			μF

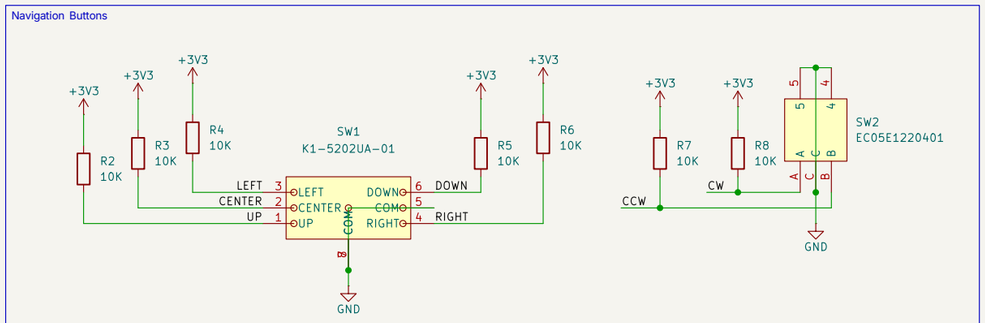
RTC



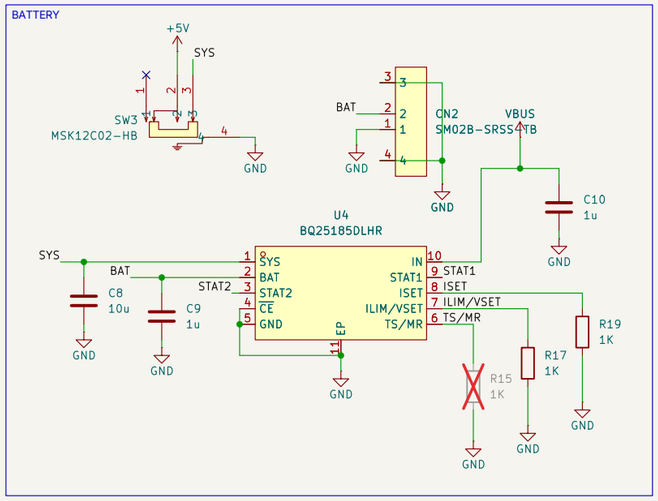
LDO



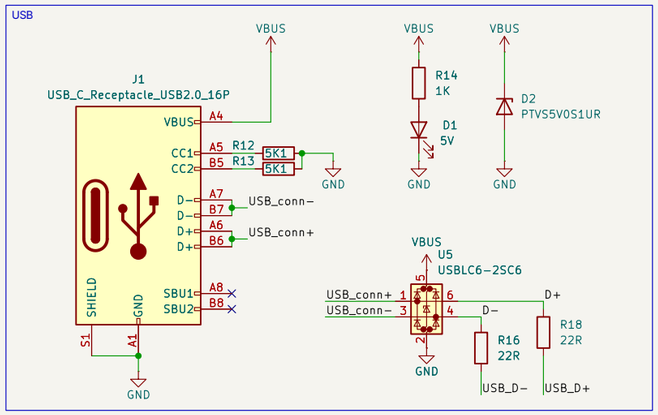
Navigation Buttons



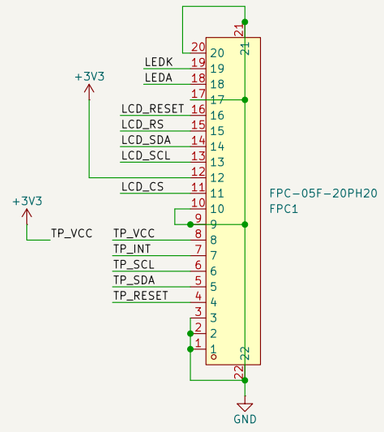
BATTERY

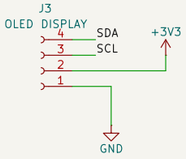


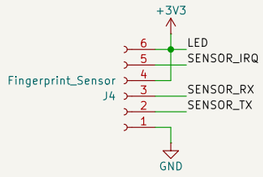
USB

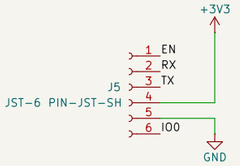


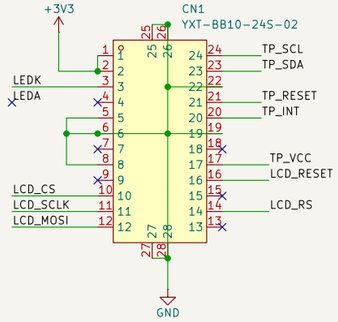


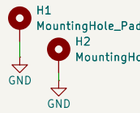












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Rev: v0.1.0